In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
from scipy.cluster.hierarchy import dendrogram, linkage, fcluster
```

In [5]:
```
df = pd.read_csv(r"E:\sales_data_sample.csv", encoding="ISO-8859-1")

print("Dataset loaded successfully!")
print("Shape:", df.shape)
print(df.head())
```

```
Dataset loaded successfully!
Shape: (2823, 25)
   ORDERNUMBER  QUANTITYORDERED  PRICEEACH  ORDERLINENUMBER    SALES  \
0        10107               30      95.70               2  2871.00
1        10121               34      81.35               5  2765.90
2        10134               41      94.74               2  3884.34
3        10145               45      83.26               6  3746.70
4        10159               49     100.00              14  5205.27

          ORDERDATE   STATUS  QTR_ID  MONTH_ID  YEAR_ID  ...  \
0   2/24/2003 0:00  Shipped       1         2     2003  ...
1    5/7/2003 0:00  Shipped       2         5     2003  ...
2    7/1/2003 0:00  Shipped       3         7     2003  ...
3   8/25/2003 0:00  Shipped       3         8     2003  ...
4  10/10/2003 0:00  Shipped       4        10     2003  ...

                    ADDRESSLINE1  ADDRESSLINE2           CITY STATE  \
0         897 Long Airport Avenue           NaN            NYC    NY
1              59 rue de l'Abbaye           NaN          Reims   NaN
2  27 rue du Colonel Pierre Avia           NaN          Paris   NaN
3              78934 Hillside Dr.           NaN       Pasadena    CA
4                 7734 Strong St.           NaN  San Francisco    CA

  POSTALCODE COUNTRY TERRITORY CONTACTLASTNAME CONTACTFIRSTNAME DEALSIZE
0      10022     USA       NaN             Yu             Kwai    Small
1      51100  France      EMEA        Henriot             Paul    Small
2      75508  France      EMEA       Da Cunha           Daniel   Medium
3      90003     USA       NaN          Young            Julie   Medium
4        NaN     USA       NaN          Brown            Julie   Medium

[5 rows x 25 columns]
```

In [6]:
```
numeric_cols = df.select_dtypes(include=[np.number]).columns.tolist()
X = df[numeric_cols]

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

print("\nNumeric columns used for clustering:", numeric_cols)
```

```
wcss = []
for k in range(1, 11):
    kmeans = KMeans(n_clusters=k, init='k-means++', random_state=42)
    kmeans.fit(X_scaled)
    wcss.append(kmeans.inertia_)

plt.figure(figsize=(6,4))
plt.plot(range(1, 11), wcss, marker='o')
plt.title('Elbow Method For Optimal k')
plt.xlabel('Number of clusters')
plt.ylabel('WCSS')
plt.show()

optimal_k = 4
kmeans = KMeans(n_clusters=optimal_k, init='k-means++', random_state=42)
kmeans_labels = kmeans.fit_predict(X_scaled)

df['KMeans_Cluster'] = kmeans_labels
print("\nK-Means clustering done. Sample results:")
print(df.head())

plt.figure(figsize=(6,4))
sns.scatterplot(x=X_scaled[:,0], y=X_scaled[:,1], hue=kmeans_labels, palette='Set2'
plt.title(f'K-Means Clusters (k={optimal_k})')
plt.xlabel(numeric_cols[0])
plt.ylabel(numeric_cols[1])
plt.show()

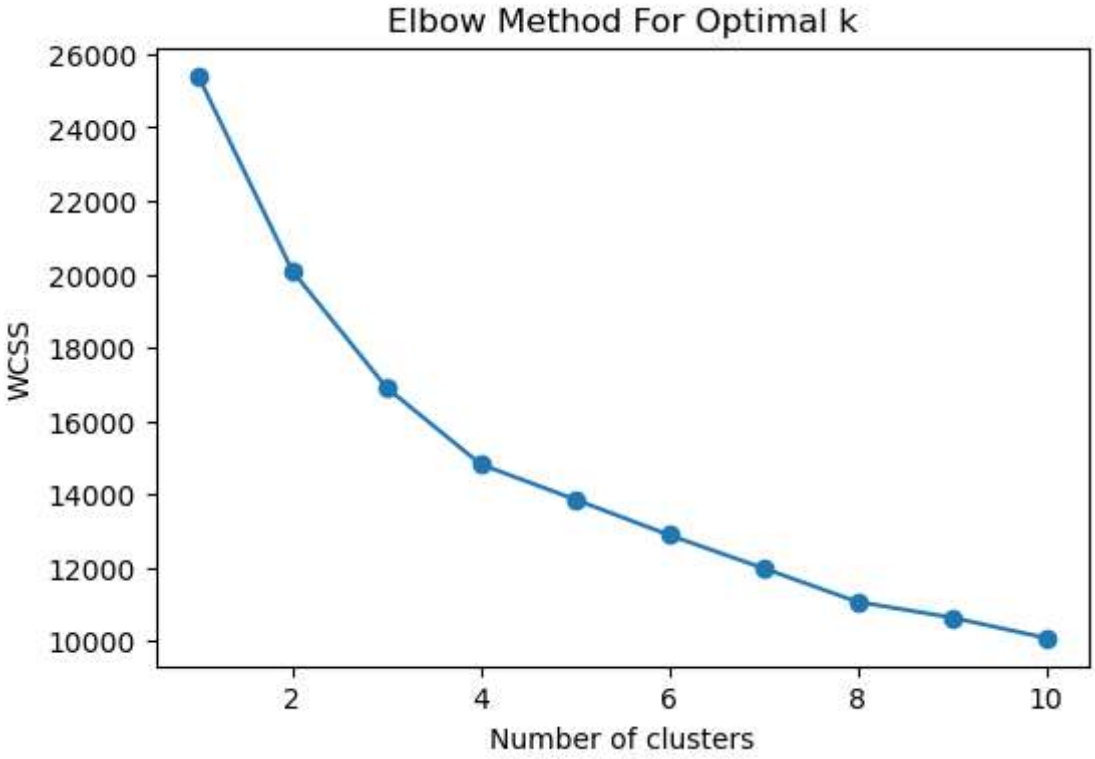
linked = linkage(X_scaled, method='ward')

plt.figure(figsize=(10, 5))
dendrogram(linked,
           orientation='top',
           distance_sort='descending',
           show_leaf_counts=False)
plt.title('Hierarchical Clustering Dendrogram')
plt.xlabel('Samples')
plt.ylabel('Euclidean Distance')
plt.show()

hc_labels = fcluster(linked, t=optimal_k, criterion='maxclust')
df['Hierarchical_Cluster'] = hc_labels - 1
print("\nHierarchical clustering done. Sample results:")
print(df.head())

df.to_csv("sales_data_clusters.csv", index=False)
print("\n Clustered data saved as 'sales_data_clusters.csv'")
```

```
Numeric columns used for clustering: ['ORDERNUMBER', 'QUANTITYORDERED', 'PRICEEACH',
'ORDERLINENUMBER', 'SALES', 'QTR_ID', 'MONTH_ID', 'YEAR_ID', 'MSRP']
```



K-Means clustering done. Sample results:

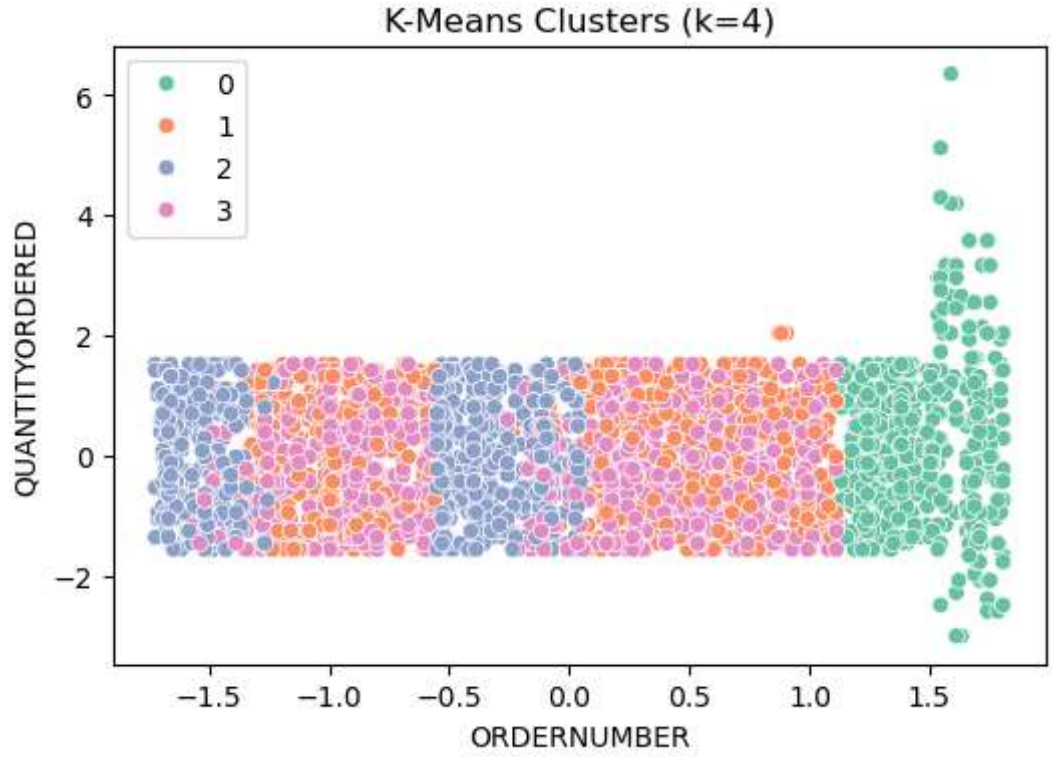
	ORDERNUMBER	QUANTITYORDERED	PRICEEACH	ORDERLINENUMBER	SALES	\
0	10107	30	95.70	2	2871.00	
1	10121	34	81.35	5	2765.90	
2	10134	41	94.74	2	3884.34	
3	10145	45	83.26	6	3746.70	
4	10159	49	100.00	14	5205.27	

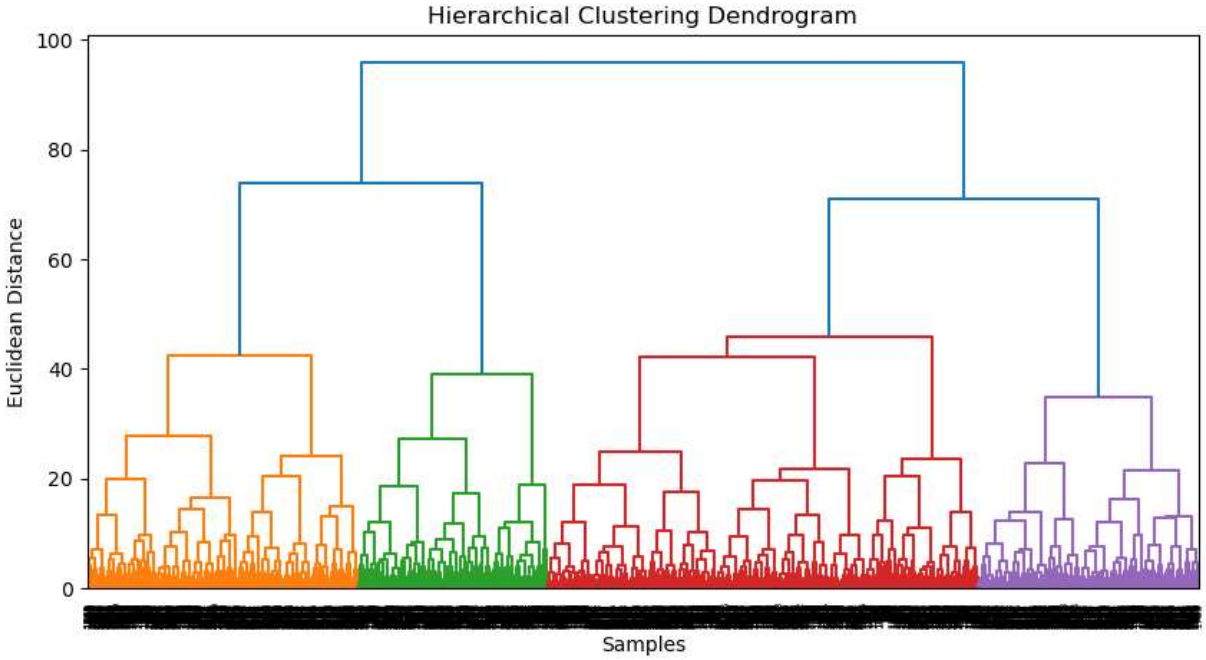
	ORDERDATE	STATUS	QTR_ID	MONTH_ID	YEAR_ID	...	ADDRESSLINE2	\
0	2/24/2003 0:00	Shipped	1	2	2003	...	NaN	
1	5/7/2003 0:00	Shipped	2	5	2003	...	NaN	
2	7/1/2003 0:00	Shipped	3	7	2003	...	NaN	
3	8/25/2003 0:00	Shipped	3	8	2003	...	NaN	
4	10/10/2003 0:00	Shipped	4	10	2003	...	NaN	

	CITY	STATE	POSTALCODE	COUNTRY	TERRITORY	CONTACTLASTNAME	\
0	NYC	NY	10022	USA	NaN	Yu	
1	Reims	NaN	51100	France	EMEA	Henriot	
2	Paris	NaN	75508	France	EMEA	Da Cunha	
3	Pasadena	CA	90003	USA	NaN	Young	
4	San Francisco	CA	NaN	USA	NaN	Brown	

	CONTACTFIRSTNAME	DEALSIZE	KMeans_Cluster
0	Kwai	Small	2
1	Paul	Small	2
2	Daniel	Medium	2
3	Julie	Medium	1
4	Julie	Medium	1

[5 rows x 26 columns]


K-Means Clusters (k=4)
0
1
2
3
QUANTITYORDERED
ORDERNUMBER
6
4
2
0
−2
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5


Hierarchical Clustering Dendrogram
Euclidean Distance
Samples
100
80
60
40
20
0

```
Hierarchical clustering done. Sample results:
   ORDERNUMBER  QUANTITYORDERED  PRICEEACH  ORDERLINENUMBER    SALES  \
0        10107               30      95.70                2  2871.00
1        10121               34      81.35                5  2765.90
2        10134               41      94.74                2  3884.34
3        10145               45      83.26                6  3746.70
4        10159               49     100.00               14  5205.27

          ORDERDATE   STATUS  QTR_ID  MONTH_ID  YEAR_ID  ...           CITY  \
0   2/24/2003 0:00  Shipped       1         2     2003  ...            NYC
1    5/7/2003 0:00  Shipped       2         5     2003  ...          Reims
2    7/1/2003 0:00  Shipped       3         7     2003  ...          Paris
3   8/25/2003 0:00  Shipped       3         8     2003  ...       Pasadena
4  10/10/2003 0:00  Shipped       4        10     2003  ...  San Francisco

   STATE POSTALCODE COUNTRY TERRITORY CONTACTLASTNAME CONTACTFIRSTNAME  \
0     NY      10022     USA       NaN              Yu             Kwai
1    NaN      51100  France      EMEA         Henriot             Paul
2    NaN      75508  France      EMEA        Da Cunha           Daniel
3     CA      90003     USA       NaN           Young            Julie
4     CA        NaN     USA       NaN           Brown            Julie

  DEALSIZE KMeans_Cluster Hierarchical_Cluster
0    Small              2                    3
1    Small              2                    3
2   Medium              2                    1
3   Medium              1                    0
4   Medium              1                    0

[5 rows x 27 columns]

 Clustered data saved as 'sales_data_clusters.csv'
```

In [ ]: